YIFEI SUN

Principal Software Engineer · Backend Platform Infrastructure & Distributed Systems · 13 Years

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SUMMARY

Principal-level backend & platform engineer with 13 years building fault-tolerant distributed systems and high-traffic platform infrastructure. Led the redesign of a critical API orchestration layer serving millions of daily requests, reducing p99 latency 5× and achieving 99.9% availability across 10+ teams. Built correctness verification frameworks, event-driven architectures (Kafka, CQRS), and observability standards adopted org-wide. Track record of building platform infrastructure that multiple teams depend on, driving cross-functional alignment across departments, and delivering complex 0→1 systems from inception through production.

LEADERSHIP HIGHLIGHTS

- Led cross-org platform modernization spanning 10+ engineering teams across 3 departments with measurable reliability impact
- Designed event-driven architectures (Kafka, CQRS) and observability standards (distributed tracing, SLO-based latency budgets)
- Built correctness verification framework — automated validation at 50% production traffic for safe, data-driven migrations
- Mentored senior engineers toward Principal-level scope; drove architectural standards via RFCs and design docs

Technologies: Java, Spring Boot, Python, Node.js · Service Orchestration, API Design, Fault Tolerance, Idempotency · AWS, Kubernetes, Docker, Kafka, Redis, PostgreSQL · GraphQL (Netflix DGS), Microservices, Event-Driven, CQRS · Datadog, Grafana, Distributed Tracing · Keycloak, SSO, OAuth 2.0

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead

Oct 2019 – Present

High-Traffic Platform Modernization — Orchestration & Fault Tolerance

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core platform behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions).
- Introduced a new orchestration service consolidating 12 downstream integrations into 5 well-defined services with circuit breaking, fault isolation, and idempotent retry — defining the platform infrastructure that teams across the org depend on.
- Produced initial architecture in ~2 weeks; shipped first milestone in 6 months; completed ahead of schedule.

p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)

GraphQL Federation — Unified Serving Layer

- Designed and integrated a Netflix DGS subgraph into CarGurus' federated GraphQL gateway, creating a unified API layer for frontend, backend, and AI agent consumers from a single graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies; built load tests and staging integration tests for production-grade reliability.

Digital Commerce Platform — 0→1 Under Ambiguity

- Led backend design and launch of multiple greenfield products using microservice and event-driven architecture; enabled the company to rapidly test, launch, and safely retire product experiments without destabilizing core systems.

Parity Testing Framework — Correctness Verification at Scale

- Built an automated correctness verification framework that validated new system output against legacy at 50% production traffic — catching data integrity issues before they reached users and enabling safe, data-driven migration rollout.

Cross-Org Platform Leadership & Technical Direction

- Defined architectural standards — API design, caching, distributed orchestration, observability — adopted across multiple teams via RFCs and design docs.
- Mentored senior engineers on system design, performance, and path to Principal-level scope.

AI Integration — MCP Server & GenAI Platform

- Sole backend engineer for an MCP server on the OpenAI Apps SDK, exposing platform capabilities as tools callable by AI agents. Separately built a GenAI automation platform with multi-provider LLM orchestration, RAG-augmented inference, and an evaluation harness.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Platform

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Platform Initiative

- Drove design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration — Customer Support & Workflow Automation

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability and rollback paths.

Toast Inc. — Senior Software Engineer Sep 2014 – Oct 2017

Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth; designed and implemented core backend services (Order Service, Labor Service, foundational APIs) that became the platform other teams built upon.
- Built initial AWS deployment pipelines and automated testing framework, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners, ensuring reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010